

Personal Research Ontologies for Personal Open Science

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Abstract

As researchers conduct their work, they incrementally construct a *personal research ontology*: a rich, personal understanding of their field and its phenomena. Yet most of this knowledge remains internalized, with only fragments surfacing in publications. This position paper argues that research is fundamentally an act of personal ontology construction and proposes *Personal Open Science* (POS), a lightweight method for externalizing a researcher's personal research ontology for downstream use. The method requires no machine learning or language models, and it centers on the researcher rather than on the publication, the traditional focal unit of Open Science. Building on daily reading habits and a minimal annotation scheme, the approach adds little overhead while producing a structured, explorable representation of a researcher's interests, influences, and expertise. I demonstrate the method through my own practice of annotating over 45,000 documents across a decade of research. I discuss how externalized personal research ontologies could serve multiple downstream purposes, such as the discovery of collaborators and self-reflection, by revealing the gap between a researcher's understanding and the shared ontology of their field. When many researchers externalize their ontologies, the resulting network of overlapping and complementary knowledge structures becomes a resource for collective intelligence, making visible the distributed expertise of a research community and enabling new forms of intellectual coordination across disciplinary boundaries.

CCS Concepts

• **Computing methodologies** → **Ontology engineering**; • **Human-centered computing** → *Collaborative and social computing theory, concepts and paradigms*; • **Applied computing** → Document management and text processing.

Keywords

Open Science, knowledge externalization, collective intelligence, scholarly communication

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1 Introduction

An ontology is an explicit specification of a conceptualization.

—Tom Gruber, 1993

This position paper starts from a simple observation: research is fundamentally an act of personal ontology construction. The term *ontology* originates in philosophy, where it denotes the study of what exists. In computer science, the term was adopted to mean an explicit specification of a conceptualization [21, 22]. Every researcher constructs a conceptualization through daily practice, by reading scientific articles, sorting findings into categories, observing connections and patterns, and identifying open research questions. I call this evolving conceptualization a **personal research ontology**. The personal research ontology is the conceptual structure through which a researcher makes sense of a field: the topics they track, the connections they draw among them, and the relative significance they attach to different ideas and findings. Most of this personal ontology remains implicit, locked inside the researcher's head. This opacity differs from tacit knowledge (i.e., the non-codified know-how acquired through informal socialization) in organizational theory [25]. A personal research ontology is personal knowledge [42]. This knowledge is, in principle, codifiable: it is the conceptual structure that is shaped by how a researcher engages in sense-making with respect to the scientific literature and their discipline, how a researcher selects problems, interprets evidence, and navigates their field. This structure is meaningful, but in practice, it is not directly observable. In the qualitative research tradition, the accumulation of internalized experiences is valued as personal and professional growth [13, 30]. But internalized knowledge is fragile: we tend to forget, or we simply move on to other topics. And the personal experience cannot be searched, shared, or built upon by others.

Open Science initiatives have made progress in making research outputs visible and accessible [11, 14, 15]. Executable notebooks [48] and research objects [6, 45] were once heralded as successors of the scientific paper. Yet, while preprints, datasets, and code repositories are now openly available, current publishing remains centered on the publication as the primary unit of scientific communication. A researcher's Google Scholar profile, for instance, lists their published works but reveals nothing about the thousands of papers they intellectually engaged with in the process of 'generating' these works, the topics they explored but never published on, their latest (unpublished) research interests, or how they organize their understanding of the field. Thus, a researcher's publications provide only a narrow and often outdated window into the researcher's personal research ontology.



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Meanwhile, the landscape of scientific production is shifting. With generative artificial intelligence, the marginal cost of producing paper-shaped research output is collapsing to zero [8, 27]. Generative AI is diminishing the reliability of publication-based indicators. Furthermore, traditional measures of research impact, such as the h-index, cannot distinguish deep expertise from surface-level familiarity, and contemporary co-authorship practices further dilute the signal [1, 29, 35]. New ways of making research expertise visible are needed.

This paper makes three contributions. First, it introduces the concept of the **personal research ontology** and argues that constructing such an ontology is the core intellectual work of researchers. Second, the paper proposes **Personal Open Science (POS)**, a vision and a lightweight method for externalizing the personal research ontology. The method places the researcher, rather than the publication, at the center of Open Science. I demonstrate the method through my own practice of annotating over 45,000 documents across a decade of research. Third, the paper discusses what becomes possible when many researchers externalize their ontologies: a network of overlapping knowledge structures that makes the distributed expertise of a research community visible and supports new forms of collective intelligence.

2 Background

2.1 Open Science and the Primacy of the Publication

The central tenet of Open Science is to make scientific research and its dissemination accessible to all levels of society [33]. The movement encompasses far more than free access to scientific articles. Fecher and Friesike identify five schools of thought within Open Science [17], ranging from opening the research process itself to measuring impact in new ways. Initiatives such as the European Open Science Cloud aim to provide federated environments where researchers can publish, find, and reuse data, tools, and services [16], and journals increasingly adopt guidelines that promote transparency and reproducibility [39, 53].

In practice, Open Science is frequently conflated with Open Access. This reduces a multi-faceted concept to a single binary dimension: whether to publish in an open access format or not. Researchers who follow the movement typically upload preprints to repositories [54], publish datasets and code, and write open peer reviews. These artifacts orbit the same focal unit: the scientific publication, a practice dating back to the emergence of the academic journal in the 17th century. A gap persists between the grand vision of Open Science and what researchers do [38].

This publication-centrism reflects a broader tension between object-centric and people-centric views of knowledge work [2, 19], and it has a practical cost: because journal articles may take two to three years to be accepted, a researcher's profile often reflects neither current work nor latest interests. The researcher receives insufficient attention in Open Science, even though academia has a culture of professional openness rare in other sectors.

2.2 Making Hidden Work Visible

Some researchers have tried to make the hidden work in science visible. One approach is the “CV of failures” (or “shadow CV”),

which lists a researcher's rejected papers, failed funding bids, and unsuccessful fellowship applications. The practice opposes the bias of only reporting successes [49], but has not found widespread adoption. To my knowledge, only four researchers in the fields of Computing and Human-Computer Interaction have published such documents: Jessica Vitak,¹ Johannes Haushofer,² Melanie Stefan,³ and Veronika Cheplygina.⁴ Maintaining such a list requires effort without offering much in return, which is why this trend was short-lived. The lesson matters for any proposal in this space: a practice of making hidden work visible must be nearly free to sustain yet add value to the person who maintains the practice.

More systematic efforts have come from the publishing world. “Enhanced publications” were a trending topic about 15 years ago promising to link articles to their underlying data and other research objects in digital repositories [58]. “Publishing 3.0” [55] and interactive, computational papers [48] reimagined the article itself. But these efforts found traction with only a few publishers. Because these efforts depend on publishers, researchers have contributed little to their uptake. More importantly, they keep the publication as their focal point. Research information systems, such as VIVO [10] and Harvard Catalyst [23], move toward researcher-centered infrastructure by aggregating profiles for networking [3], but they populate these profiles primarily from the publication record. What a researcher explores, values, and knows remains invisible in all of these approaches.

The closest predecessor to the approach proposed in this paper is Luhmann's *Zettelkasten*, a slip-box system in which the sociologist externalized his personal knowledge structure as a hierarchy of handwritten, cross-referenced notes [46]. His practice demonstrated that making one's conceptual organization explicit can produce new insights and connections, not merely archive existing ones. Yet even Luhmann's system required writing, numbering, and cross-referencing each note by hand, an investment that few researchers sustain at scale.

A parallel practice has developed largely outside academia. Since 2008, the Quantified Self movement [51] has brought together individuals who use empirical methods to answer questions about their own lives, a practice its proponents call *personal science* [57]. These practitioners study themselves rather than a scientific field, yet they face the problem this paper starts from: each accumulates situated knowledge that is valuable to them and largely illegible to anyone else. When the community tried to pool that knowledge, the difficulty surfaced. Kloppenborg et al. co-designed the Personal Science Wiki with practitioners and found in a user study that people organized the same material in different ways, and that one person often placed a single item in several categories at once [28]. The design settled on multiple paths through the material rather than one hierarchy. Making personal knowledge collective is, in part, a problem of classification.

¹<https://web.archive.org/web/20240715184947/https://jessicavitak.com/2016/04/30/embracing-failure-heres-my-cv-of-academic-rejections/>

²https://web.archive.org/web/20240324054716/https://crlte.engin.umich.edu/wp-content/uploads/sites/5/2020/05/Johannes_Haushofer_CV_of_Failures.pdf

³<https://www.nature.com/articles/nj7322-467a>

⁴<https://web.archive.org/web/20250908200410/https://veronikach.com/failure/shadow-cv/>

2.3 Folksonomies and Social Tagging

The idea that individuals can externalize how they classify what they encounter online is not new. In the mid-2000s, a wave of Web 2.0 social bookmarking and social tagging services [31, 40] let users label web pages, photographs, and references with free-form keywords. Services such as del.icio.us and Flickr popularized the practice, and the bibliographic services CiteULike, BibSonomy, and Mendeley brought it to scientific literature, letting researchers tag and share the papers they read [24]. The aggregate of these user-assigned tags was called a *folksonomy*, a classification that emerges bottom-up from many individual acts rather than being imposed top-down by experts [20, 32]. A lively debate followed about whether such bottom-up tagging could substitute for, or only complement, the formal ontologies of knowledge engineering [36, 47].

The fate of this wave is instructive. Some of these services did not survive. CiteULike shut down in 2019, and social bookmarking for science never reached the scale its proponents imagined. The signal that Mendeley readership counts provided was studied as an early indicator in the science of science [37, 43], yet the practice of tagging imposed an overhead that most researchers eventually abandoned. The reading behavior of scientists carries a useful signal, but capturing it has proven difficult to sustain, a lesson this paper returns to in Section 5.6. The underlying problem was one of incentives: tagging was extra work performed primarily for the benefit of the collective, not for the individual researcher. When the collective benefit failed to materialize at scale, the individual effort lost its justification.

2.4 Ontologies and Collective Intelligence

In knowledge engineering [50] and the field of the Semantic Web [7, 52], ontologies were developed as formal, explicit specifications of a shared conceptualization in a domain of interest [50], designed so that knowledge could be exchanged between systems and reused [21]. A scientific field can be viewed in the same way. The field's taxonomy of topics, its canonical references, and its sense of which problems matter form a shared ontology that members of the field carry, imperfectly and partially, in the public record of publications and in their heads.

This view connects directly to collective intelligence. Groups perform well when they develop a shared understanding of who knows what – a phenomenon called transactive memory [56]. In science, this capability is weak. Scientists self-organize into teams, but team formation comes at a considerable cost. A field experiment at Harvard Medical School showed that even small search costs affect who collaborates with whom [9]. The information needed to identify good collaborators, namely researchers with deep knowledge of specific topics, exists, but it is distributed across thousands of heads and only weakly signaled by publication records. Making personal ontologies explicit is, in this framing, a way of building a transactive memory system for science.

3 From Implicit Knowledge to Personal Open Science

3.1 Personal Research Ontologies

Constructing a personal research ontology is not incidental to research. It is what much of research is about. Reading a paper means deciding where it fits in a broader knowledge structure. Conducting a literature review means restructuring a region of the ontology. Identifying a research gap means noticing a place where the ontology is thin.

Three properties matter for this paper. First, the ontology is *personal*. It is shaped by the researcher's interests, training, personality, curiosity, intellectual capacity, and environment (in the sense of what Rhodes called "press" in his four-P model of creativity [44]). Two researchers in the same field will organize their ontologies differently, and those differences carry meaning. Second, the ontology is *evolving*. It changes with every paper read and every project completed. At any moment it reflects the researcher's current understanding, unlike the publication record, which trails years behind. Third, the ontology is *broader than the publication record*. It covers everything the researcher has engaged with, including topics they explored but never published on. The published record samples only a thin slice of it.

Each personal ontology relates to the shared ontology of the researcher's field. A personal ontology is a partial and idiosyncratic projection of this shared structure. The overlap between the two indicates how well the researcher has internalized the field. The divergence indicates either a gap in learning or, sometimes, an original perspective on topics the field has not yet charted. This relation grounds the use of personal ontologies as building blocks of collective intelligence (see Section 5) and for self-assessment (Section 5.3).

The problem is that this personal research ontology is implicit. The next section proposes a method that externalizes a useful approximation as a by-product of daily work. The externalization need not capture the full ontology to be useful. Even a partial, structured trace of the researcher's engagement with the literature carries information that the publication record does not.

3.2 Personal Open Science: A Vision and a Method

Personal Open Science (POS) is both a vision and a lightweight method. The POS vision holds that Open Science should extend beyond published artifacts. Traditional Open Science makes publications and their associated artifacts openly available. Personal Open Science complements this by making visible the broader knowledge structure that a researcher builds through daily practice. The POS method is one instantiation of this vision: a deliberately lightweight approach combining a long-term reading habit with a minimal annotation scheme. Everything else, such as the visualizations and the question-answering interface described in Section 4, is a downstream use of the annotated corpus. This section describes the method as I have practiced it since 2016. It is an existence proof, not the only possible instantiation. Because the method centers on reading and curation, it captures primarily one dimension of

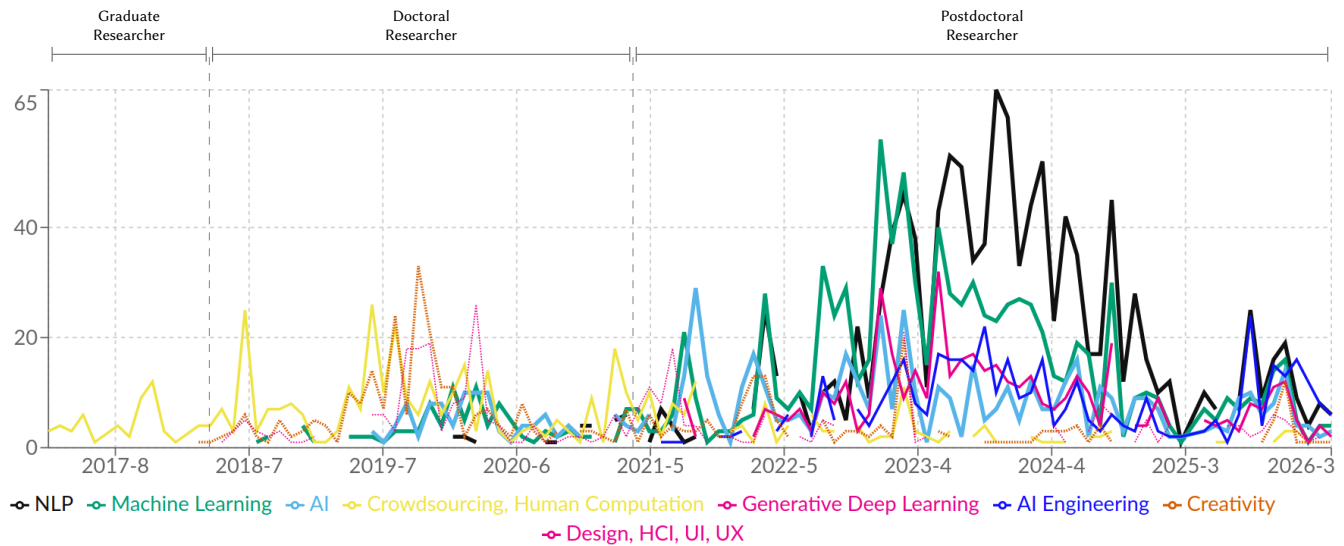


Figure 1: Reading activity over time across selected top-level topics, derived from timestamps of documents rated as relevant or important (“!” prefix or higher) in the annotated collection. Shifts in attention, such as the rise of generative AI topics from 2022 onward, become visible without any manual tracking. The brackets above the plot mark three career stages: graduate researcher until April 2018, doctoral researcher until April 2021, and postdoctoral researcher thereafter.

the personal research ontology. Future extensions could externalize other dimensions, such as research activities, methodological decisions, and collaborative interactions, to approximate the full personal research ontology more closely.

3.2.1 Design goals. The method is designed around a simple set of goals:

- (D1) Integration into the daily workflow,
- (D2) Low effort and friction,
- (D3) Value added, and
- (D4) Support for exploration and self-reflection.

The first two goals follow from the lesson of Section 2.2, and together the design goals respond to the incentive failure of folksonomies (Section 2.3): where social tagging asked researchers to do extra work for a collective payoff that might never arrive, POS embeds the annotation into the researcher’s daily work and delivers individual value from day one (D3; see Section 4.1). A method for Personal Open Science becomes more useful the longer it is sustained. The accumulated data must provide more value than the individual pieces alone (D3), and the structure the method produces should allow the researcher to explore and reflect on it (D4).

3.2.2 Annotation scheme and consistent habits. The core of the method is a habit I developed during my Master’s degree and solidified during my first graduate research position. Some researchers manage their literature by writing detailed notes on each paper they read, for instance in notebooks, spreadsheets, or annotated bibliographies. Reference managers such as Zotero organize bibliographic metadata, and personal knowledge management tools such as Obsidian, Roam, and Notion offer environments for linking notes and ideas in the tradition of Luhmann’s *Zettelkasten*. These

practices produce valuable records of engagement with the literature, but they impose a significant recurring cost on daily work. A few minutes of note-taking or curation per paper accumulate to hours each week. A practice intended to run for years must be lighter than that (D1, D2). The method proposed here is not another tool. It requires no separate applications, only the file system every researcher already uses. The method supports continuous foraging of information with minimal overhead.

The method’s premise is that anything I read may become relevant later. By consistently applying the method, I configure my informational environment for future serendipitous discovery. Thus, the first element of the method is a habit related to storage: I save everything I read as PDF documents on my computer. Whether it is a scientific paper, a Wikipedia article, a blog post, or a BBC news article, every document enters the same archive. If needed, optical character recognition (OCR) is directly applied in my PDF reader before saving the document. The archived PDFs serve as the raw material for all downstream uses described in Section 4.

The second element is a lightweight annotation. When saving a document, after reading or skimming it, I assign a coarse importance score encoded in the filename (see Table 1).

I deliberately do not distinguish between relevance and importance. A literature review on a topic outside my current interests is still important and may become relevant in future research. I intentionally do not keep notes or write summaries of articles. I started this habit in 2016 with the vision of one day querying the full text of the documents with language models, which would make manually written summaries redundant.

The method’s third element is a hierarchical folder structure. The top level distinguishes three areas: “Literature” (the most relevant material, accessed daily), “Academic Working” (literature

Table 1: The annotation scheme. The importance score is encoded as a prefix in the filename, which keeps the annotation visible in every tool that lists files. The remainder of the filename is left as the publisher assigned it.

Prefix	Meaning	Example filename
(none)	not yet classified	2108.13161.pdf
-	not relevant or important	-1810141115.full.pdf
!	relevant or important	!1478744.pdf
!!	very relevant or important	!!782780.pdf
!!!	extremely relevant or important	!!!p1245-yu.pdf

about science itself), and “Miscellaneous” (less relevant material). Below that, folders represent topics and may nest to arbitrary depth. For instance, the folder “Creativity” contains a folder “Creativity Support Tools,” which in turn contains other folders. If there are synonyms for a topic, a comma in the folder name separates them. Three reserved folder names recur throughout the folder hierarchy and encode consistent semantics. “Applications” collects articles on how a method or technology is applied. “Issues” collects open problems and challenges of a topic. “General, Theory” collects general information on fundamental concepts of a topic.

This is the entire annotation scheme: an importance prefix and a place in a topic hierarchy, assigned at the moment of saving a file. The marginal effort per document is a few seconds (D1, D2). After a decade of practice, the structure comprises 45,444 PDF documents in 28,075 folders, with a maximum folder depth of 16. The folder hierarchy, the importance prefixes, and the file timestamps together form an externalized approximation of one part of my personal research ontology: what topics I distinguish, how I relate them, which documents I attach to them, how much I value each document, and when I engaged with it.

Formally, the folder hierarchy is an annotated taxonomy, not an ontology in Gruber’s sense. The knowledge structure is deliberately designed as a simple ontology [34]. The personal research ontology, as defined in Section 3.1, should be considered as the complete conceptual structure in the researcher’s head: the topics, the connections drawn across them, the methodological preferences, the sense of which problems matter. The folder hierarchy externalizes one facet of this structure: the topics the researcher distinguishes, their nesting, and a coarse importance judgment on each document. Other facets remain implicit, and the gap between the full ontology and its externalized approximation is where future work lies. For the uses proposed in this paper, a lightweight structure that a researcher maintains is more useful than a formal ontology that nobody builds.

4 Downstream Uses: What the Externalized Ontology Enables for the Individual

The method described above produces a document corpus organized into a topic hierarchy, rated by importance. This section describes what I have built on top of this corpus, namely a public web interface that visualizes the ontology and a question-answering interface. These are two downstream uses, not part of the POS method. Other researchers could build different applications on the same kind

of corpus. The section outlines the value a decade of practice has shown at the individual level. I describe the individual benefits before any collective ones, because a practice must benefit the individual researcher before any collective benefit can materialize (D3). Section 5 broadens the perspective to collective intelligence.

4.1 An External Memory and Object for Self-Reflection

The hierarchical document storage functions as an external workbench onto which I offload memory. I often cannot recall what I read a few days ago, but I can find it in the list of recently read documents. A simple shell command can surface recently read documents by timestamp. When I need the most relevant literature on a specific topic, I navigate the folder hierarchy and let the importance prefixes surface the documents that matter. The full text remains searchable as a fallback. The ontology converts a vague sense of “I have read something about this” into a retrievable artifact.

Because the ontology is externalized, I can examine it as a whole. The folder structure shows me where my reading has been deep and where it has been shallow. The reading timeline shows how my interests have shifted (Figure 1). Researchers routinely describe their interests in abstract terms. An externalized ontology lets me check these descriptions against evidence about my behavior (D4).

4.2 Publishing and Visualizing the Ontology

The annotated corpus becomes Personal Open Science when parts of it are published. Periodically, I run a script that traverses the folder structure, extracts metadata, and regenerates several interactive views on the ontology. Titles and authors are parsed from the PDFs using Ai2’s Science Parse, which succeeds for about two thirds of the documents. Where parsing fails, my manual annotation of the title is used, yielding 100% paper title coverage. The last-modified date of each file indicates when I last read or annotated it. The resulting data feeds several views on my personal website (Figure 1, Figure 2, and Figure 3).

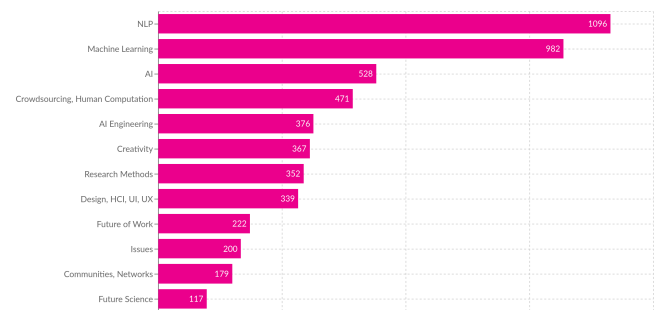


Figure 2: Screenshot of an interactive visualization of the largest first-level topics of the folder hierarchy, ranked by the number of documents rated as relevant or important (“!” prefix or higher). A view of this kind replaces the keyword list of research interests found on most academic websites with counted evidence of sustained attention.

What am I reading? The website lists the last hundred documents I most recently read. Assembling this list adds little overhead in my

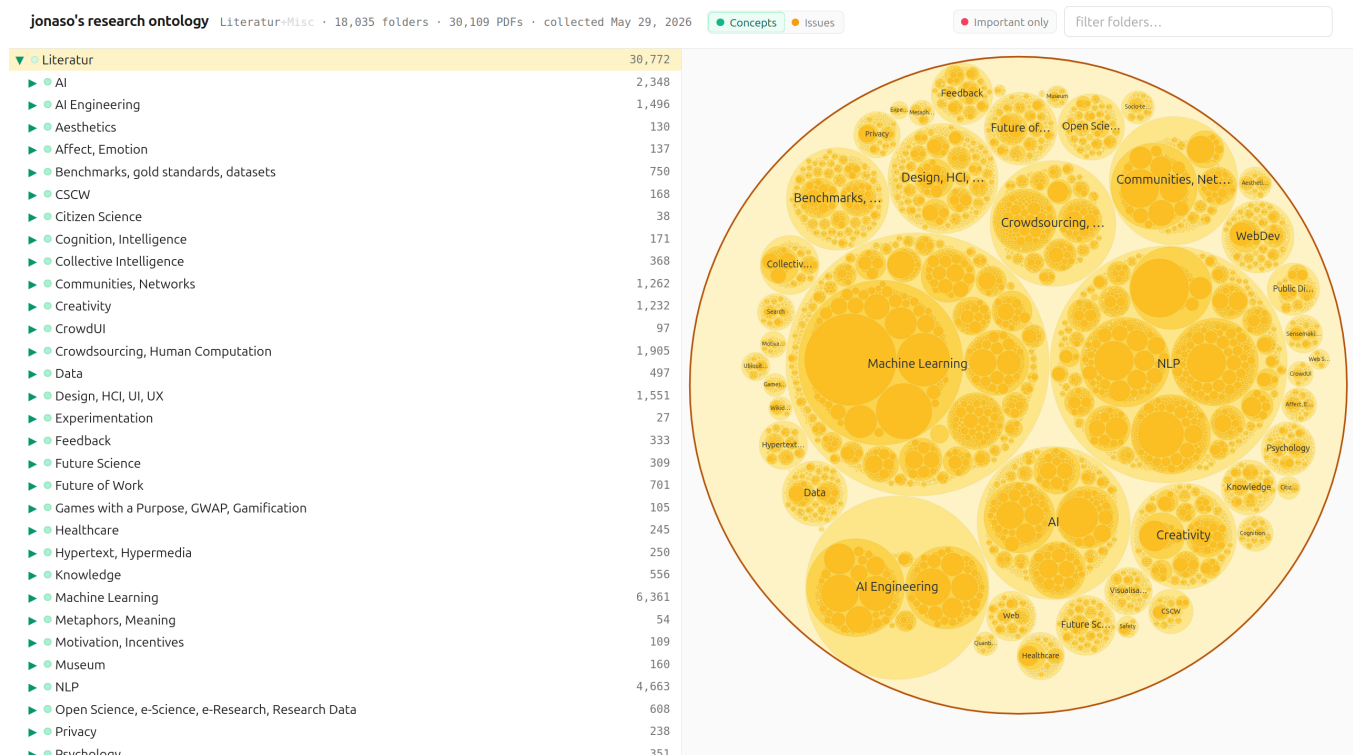


Figure 3: The public web interface to the externalized personal research ontology, showing an explorable view of research interests constructed from recently read publications.

daily work, since it falls out of the annotation habit. The reading record includes everything I engage with, not only scientific papers. For anyone deciding whether to contact me about a topic, the last hundred documents I read are a far better indicator than my last accepted paper.

What do I consider important? Visitors can explore my research interests down to the first level of the folder hierarchy. Many researchers list their interests on their websites in a keyword list. The difference here is that my interests are backed by behavioral evidence. I can demonstrate my engagement with a research topic by showing the number of documents I have read and how I rated their importance. The ontology thereby documents engagement, not only research output. It also signals engagement with topics I have never published on.

Who influences me? Based on what I read, the website surfaces the authors whose work I find inspiring and important, structured into three tiers: highly influential, influential, and relevant. Clicking on an author shows their publications in my collection and their keywords. Clicking on a keyword reveals other authors who have written about the same topic, and a switch highlights co-authors. Influence on social media is measured in followers. Here, it is measured in sustained reading attention. This view has proven valuable on specific occasions. It helped me identify external examiners and the opponent for my own PhD thesis, a task that requires knowing whose work one's research builds on. It provides candidates for new projects and collaborations.

4.3 Querying the Ontology

The annotation method requires no artificial intelligence, but the resulting corpus is well suited to processing by AI. To demonstrate this, I built a retrieval-augmented generation (RAG) interface that enables question-answering over the corpus (Figure 4). The folder hierarchy and the importance prefixes act as retrieval signals: a query about a topic can be scoped to the corresponding region of the hierarchy, and highly rated documents can be weighted above low-rated ones. A decade of lightweight manual annotation does the work that elaborate metadata schemes or model retraining would otherwise require.

Ontology engineering distinguishes terminological knowledge (concepts and their relations) from assertional knowledge (facts about instances) [5]. The current interface queries both layers: the folder hierarchy, importance ratings, and document titles, but not the full text of the documents. Extending retrieval into the document contents is a natural next step, but the structural layer alone already supports substantive answers (see the pre-formulated queries in Figure 4).

5 A Research Agenda for Collective Intelligence

The benefits described so far accrue to a single researcher. The more consequential argument, and the reason for this position paper, concerns what happens when many researchers externalize their personal research ontologies. Each ontology is a structured, machine-readable account of who knows what. Together, these

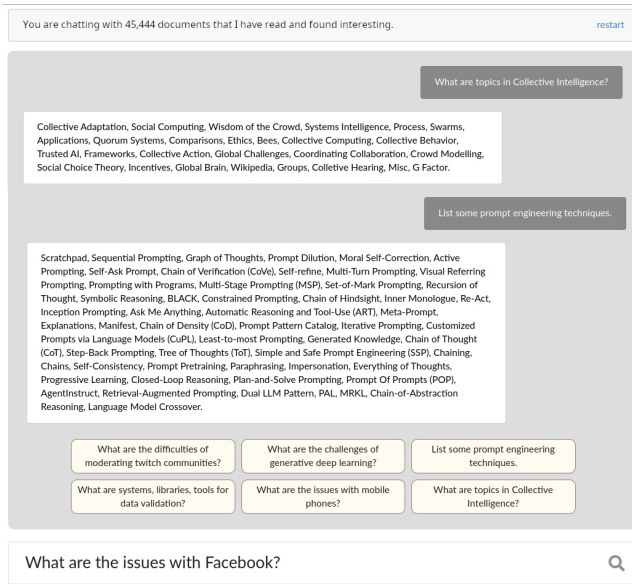


Figure 4: The retrieval-augmented question-answering interface to the personal research ontology. Visitors can converse with the folder structure, either by typing their own question or by selecting one of the suggested questions, which are derived from the topics of the underlying document collection. The interface queries the terminological structure of the ontology (folders, ratings, and titles), not the full text of the more than 45,000 annotated documents. Future versions could extend retrieval into the document contents.

ontologies would form a network of overlapping and complementary knowledge structures: a living map of a research community’s distributed expertise. This section discusses the collective benefits of making individual researchers’ ontologies visible and legible.

5.1 Collaborator Discovery and Team Formation

Today, finding a collaborator means relying on publication records, conference encounters, and word of mouth, a process whose search costs shape who collaborates with whom. The first direction concerns discovering potential collaborators based on what researchers read and value, not only on what they have published. Externalized ontologies would let a researcher find others whose ontologies overlap with theirs (shared interests) or complement theirs (adjacent expertise), including researchers who have not yet published in the area. Whether ontology overlap is a better predictor of productive collaboration than the publication and co-authorship signals used today is an open empirical question, and a study in the tradition of Boudreau et al.’s field experiment [9] could begin to answer it.

These benefits would not be distributed evenly, and that is a feature. Early-career researchers and researchers from underrepresented regions have the thinnest publication records and the weakest informal networks, yet they may have deep engagement with specific topics. A matching infrastructure built on personal ontologies could let them signal their engagement years before the

publication system certifies it. POS could democratize competence signaling in science.

5.2 Aggregate Field Ontologies

A second direction treats aggregation of many personal ontologies as a bottom-up complement to shared scientific classification schemes. Schemes such as the ACM Computing Classification System (CCS) are constructed top-down by committees and revised rarely. Unlike the flat folksonomies of the social-tagging era, aggregated hierarchical ontologies could preserve the relations between topics and the importance assigned to them, not only the labels. Where many researchers independently create similar structures, the aggregate would reveal the de facto shared ontology of a field, weighted by actual attention rather than editorial decision. Where their structures diverge, it would reveal contested or emerging conceptualizations, which are often the most interesting regions of a field. Such a living ontology would update continuously as the behavior of the community shifts, rather than at the cadence of committee revisions. Whether idiosyncratic personal structures can be aligned into a coherent aggregate is the open question here, and a feasibility probe on the ontologies of a single subfield would be a natural first test.

5.3 Assessing Individual Expertise and Learning

A third direction follows from the relation between the two kinds of ontologies. Section 3.1 described the personal ontology in the sense of Polanyi’s personal knowledge [42] as a partial projection of the field’s shared ontology. Once both sides are explicit, the comparison becomes computable. Where a researcher’s structure is deep and well organized relative to the field’s, it would signal genuine expertise. Where it is shallow, or organized in ways that diverge from the field’s conventions, it would signal an area for further learning (or, perhaps, an original perspective worth examining).

This direction has direct implications for doctoral education. Becoming an expert largely means internalizing the shared ontology of a field. The gap between a student’s personal ontology and the field’s aggregate ontology could serve as a concrete, evolving object for self-assessment and for student-supervisor conversations, making visible which regions of the field the student has engaged with deeply and which remain thin. Used this way, the personal research ontology is a formative aid that prompts reflection and dialogue, and whether it functions that way in practice is something a longitudinal study could examine.

The same comparison must not become a high-stakes evaluative instrument. Treating the gap as a metric in qualifying decisions, as a complement to or a replacement of the fallible h-index, would turn POS into a form of surveillance, and it would penalize the divergence mentioned in Section 5.2. A student whose ontology departs from the field’s conventions may have a thin understanding, but may equally be cultivating an original perspective, and the gap alone cannot separate the two. The comparison should open a conversation in which student and supervisor interpret the divergence together, not automate a judgment about it.

5.4 A New Data Stream for the Science of Science

A fourth direction treats aggregated reading behavior as one data stream that science has only begun to tap. Mendeley readership counts were an early step in this direction [37], but they captured attention as flat counts. Citation data reveals attention only after a multi-year publication delay. Aggregated personal ontologies would reveal not only what a community is reading now but how it organizes and ranks what it reads. Emerging topics would appear as rapidly growing regions across many ontologies long before they appear in citation statistics. Funders and institutions could observe where the attention of a field is moving, and historians of science would gain a record of the reading behavior behind discoveries, a layer of evidence that has been almost entirely invisible. Whether reading-derived signals anticipate citation-based ones is testable, for instance by checking such signals retrospectively against the emergence of topics already visible in the citation record.

5.5 A Substrate for Distributed and Extended Cognition

The directions described so far treat externalized ontologies as data to be searched, matched, and aggregated. A fifth and more exploratory direction treats them as components of cognitive systems. Clark and Chalmers argued that cognition does not stop at the skull [12]. When a person relies on an external artifact as fluently and routinely as on biological memory, that artifact functions as part of their cognitive system. The document hierarchy meets this standard. After a decade of daily use, it holds over 45,000 documents organized into more than 28,000 folders, far more structure than any researcher could maintain in their head. The externalized ontology is therefore not a copy, but an extension of a mental structure – a workbench on which a researcher can hold and manipulate more of a field than unaided memory allows.

Distributed cognition extends this view from individuals to systems of people and artifacts. Hutchins showed that cognitive accomplishments, such as navigating a ship, are often properties of such systems rather than of any single mind within them [26]. Science may be the largest distributed cognitive system humans have built, yet the coupling between its minds runs almost entirely through publications, a channel that is slow and lossy. Published personal ontologies would offer a higher-bandwidth interface between researchers. A colleague could inspect not only what I concluded in the past but how I structure a field, which documents I attach to each topic, and where my understanding is deep or thin. Reading structure, rather than only reading conclusions, lets one mind borrow the organization of another.

The cognitive systems built on this substrate need not contain only humans. Because the externalized ontology is machine-readable, it can ground AI systems in a specific researcher's knowledge and values, as Section 4.3 demonstrates. One possibility at the collective scale is a layer of AI systems with access to many published ontologies. Such systems could route a question to the researchers best equipped to answer it, identify complementary expertise when assembling a team, or synthesize how different researchers organize the same topic. The collective intelligence of science would then reside neither in human heads nor in machines

alone, but in a distributed system that couples both through explicit, shared representations of who knows what.

5.6 Open Questions

Two questions cut across the directions above and shape their feasibility.

Adoption. The lesson of the “CV of failures” is that visibility practices die without immediate personal value. The same lesson emerged from the social bookmarking services that once tried to crowdsource classification and have since largely disappeared (cf. Section 2.3). POS is designed so that the individual benefits accrue from day one (cf. Section 4) and the collective benefits require no coordination to begin. This inverts the folksonomy incentive structure: adopting a social tagging service was a leap of faith that paid off only at scale, whereas a researcher who maintains a personal ontology gains an external memory and a queryable corpus whether or not anyone else participates. The collective layer is a bonus, not the justification. Still, the network effects that make ontologies most valuable require many participants, and the path to widespread adoption is rocky.

Comparability. Personal ontologies are idiosyncratic by design. Aggregating or comparing them, which several of the directions above require, means mapping between structures that different researchers organize differently. Lightweight conventions, like the reserved folder names in my scheme, may provide enough common structure. Heavyweight standardization would defeat the purpose, because the idiosyncrasy is part of the signal. Finding the right level is an open design problem. Modern language models make this problem more tractable than it was during the Semantic Web era: approximate semantic alignment of heterogeneous hierarchies no longer requires a shared formal schema, only enough overlap for a model to detect thematic correspondence. The key is to learn from past failures, such as the over-standardization of the Semantic Web [52], and design systems that are lightweight and easy to use.

6 Limitations

Both the method and the research agenda have limitations that a position paper should state plainly.

Adoption. The method is not designed to retrospectively annotate an existing corpus of documents. This goes against the ethos of the method. Instead, it is meant to be applied in daily research practice, and benefits accumulate from the point of first use onwards. The broader collective value depends on widespread participation and researchers' willingness to make their ontologies publicly available. Whether the individual incentives will translate into the collective benefits is a question for future research.

Competence signal. Researchers with different working styles may produce ontologies of varying richness and visibility. As a result, some equally knowledgeable researchers may appear less expert because they document and organize their work less extensively. This risks rewarding documentation practices rather than underlying expertise. This work does not advocate for using personal research ontologies as an evaluative instrument (cf. Section 5.3).

Privacy and strategic disclosure. A reading record is revealing. Researchers may hesitate to disclose what they read, fearing that

competitors could infer their plans and scoop them. Or they may not want to publish their reading on sensitive topics. Keeping research secret is arguably wasteful for society [4, 41], but individual incentives do not always align with that view. POS does not require full disclosure. In my own practice, the choice of what to publish is still my own. The right granularity of disclosure remains an open design question.

Gaming. Any measure that counts becomes a target [18]. If externalized ontologies serve as signals of expertise, some will inflate them by bulk-saving documents they never read. Large language models lower the cost of generating plausible folder hierarchies, making structural depth alone an unreliable safeguard. The temporal dimension, years of consistent behavior with timestamps that track evolving interests, provides some resistance to fabrication, but it is not foolproof. This is one reason the paper frames the externalized ontology as a formative aid rather than a standalone credential. Used alongside publication records, peer assessment, and other indicators, the personal research ontology adds a dimension that is otherwise invisible. Whether the signal remains informative in the presence of deliberate gaming is an empirical question.

Partial staleness. Some limitations follow directly from the method's simplicity. A rating is assigned once, at the moment of filing, but importance or relevance is not stable. Judgments shift as a field moves on, and parts of the ontology become outdated. Categories and organizational structures that reflected my thinking in 2016 may no longer represent how I conceptualize the field today. One solution would be to gradually update the importance ratings as I revisit documents. However, the documents in the outdated regions are precisely the ones I no longer revisit. Better solutions are needed, such as a temporal decay in which a rating loses weight the longer a document goes untouched.

Generalization. This paper rests on a single sustained existence proof: my own decade of practice. Whether the method transfers to researchers with different working styles, and whether the projected collective benefits materialize, are open empirical questions for the community to study.

7 Conclusion

This paper argued that research is fundamentally an act of personal ontology construction, and that Open Science should make the researcher, not the publication, its focal unit. Personal Open Science demonstrates that externalizing a personal research ontology requires little effort. A filename convention, a folder hierarchy, and consistent small habits suffice. The externalized ontology is useful and pays the individual researcher back immediately, not only as an extension of working memory and an object of self-reflection, but also through future downstream uses that are not yet apparent. The larger opportunity, however, is collective. Science already possesses a distributed map of who knows what, but it is scattered in publications and locked inside researchers' heads. If many researchers externalized their personal ontologies, that map would become explicit: a transactive memory system for science that lowers the friction of finding collaborators, reveals the living structure of scientific fields, and makes expertise visible beyond what the publication record alone can convey. Building and studying that map is a research agenda for the collective intelligence community.

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